Forecasting, Part 1: The Forecasting Challenge

Chapter 16, §16.1–16.2 (through the weighted moving average)

Dr. Skip Crooker MBA 775 University of Nevada, Las Vegas

Thursday, October 22, 2026

Table of contents

Objectives . . . 1
The Forecasting Challenge . . . 2
What Forecasting Is . . . 3
The Components of a Time Series . . . 4
Smoothing Methods . . . 4
The Naive Forecast . . . 5
The Simple Moving Average . . . 5
Measuring Accuracy: the Mean Absolute Deviation . . . 6
The Weighted Moving Average . . . 8
Two More Yardsticks . . . 8
The Challenge Series: the CPI . . . 9
Look before you forecast . . . 9
Forecast next month . . . 10
Backtest: what the challenge will do to you . . . 11
A Second Series: Weekly Initial Claims . . . 12
Getting the Data . . . 13
FRED: the Download button . . . 13
Las Vegas visitor volume: the LVCVA workbook . . . 13
What is in the course packs . . . 14
Working with Claude . . . 14
The lab prompt . . . 14
A bad prompt and a good one . . . 14
Verify before you submit . . . 15
Have Claude argue with you . . . 15
Excel, if you want it . . . 15
Your Turn (from the textbook) . . . 16
Laboratory 16a . . . 16
Path 1 — upload the files . . . 16
Path 2 — let Claude fetch the files . . . 16
Your first submission . . . 17

Objectives

- Say what a forecast is, what a time series is, and name the four components a time series can contain.
- Produce a **naive** forecast, a **simple moving average** forecast, and a **weighted moving average** forecast, by hand and in Python.
- Measure forecast accuracy with the **mean absolute deviation** and choose between methods on that basis.

- Download a series from FRED, check it, and forecast its next value — the workflow of the semester-long forecasting challenge.
- Instruct Claude to do all of the above and verify that it did.

> Donnelly puts forecasting in Chapter 16, the last chapter of the book. We are reading part of it in Week 3, and we will come back to it twice more. Two reasons. First, forecasting is the most immediately *useful* thing in the book, and it is the one you will be asked to do at work before anyone asks you for a hypothesis test. Second, it is a skill, not a fact — it needs practice, and one week in December is not practice. So for the rest of the semester you will be submitting real forecasts of real numbers, and finding out how you did when the number is published. That is the **forecasting challenge**, and this note is the equipment you need to start.
>
> The methods in this note need no probability, no regression, no distribution theory. They need arithmetic and honesty about how well the arithmetic did.

The Forecasting Challenge

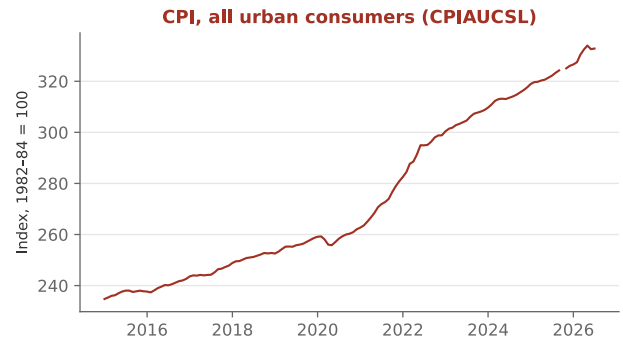
Throughout the semester you will enter forecasts in Canvas for five series. Four are on FRED, one is not.

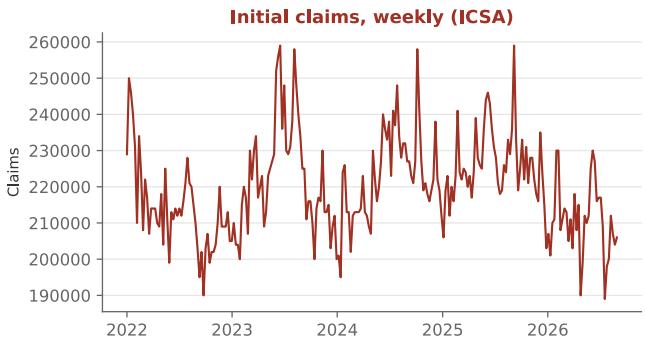
Series	FRED ID	Frequency	Published by
Consumer Price Index, all urban consumers	`CPIAUCSL`	Monthly	Bureau of Labor Statistics
Initial claims for unemployment insurance	`ICSA`	Weekly (Thursdays)	Department of Labor
New privately-owned housing units started	`HOUST`	Monthly	Census Bureau
All-transactions house price index, Nevada	`NVSTHPI`	Quarterly	FHFA
Las Vegas visitor volume	—	Monthly	LVCVA

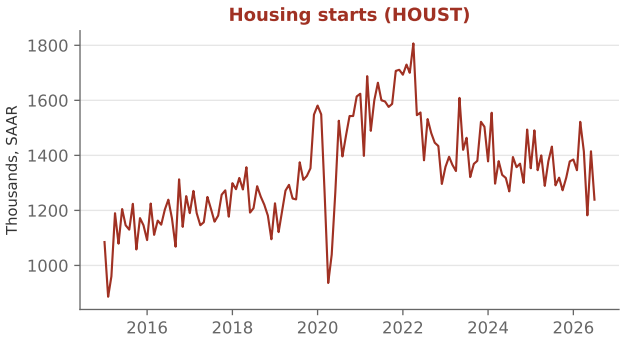
The forecast period is always *the next value the data provider will release*. Every FRED series page shows its next release date; the LVCVA posts its monthly summary about four weeks after the month ends. Due dates are on the syllabus and in Canvas.

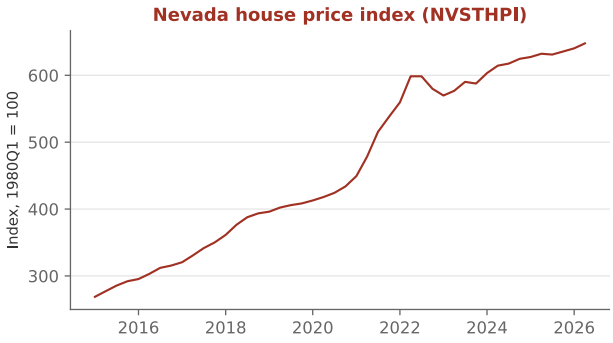
> **The rule of the challenge is the rule of the course.** Your forecast comes from a script you can rerun next period, not from a number you eyeballed off a chart or a figure Claude stated in prose. When you are asked why you submitted the number you submitted, the answer is a method, a window of data, and a measured error — all three.

Here are the five series as they stand today. Look at them before reading on. Which ones climb? Which ones jump around a level? Which one has a calendar in it?











What Forecasting Is

> **Qualitative forecasting** relies primarily on a knowledgeable person's intuition and judgment to predict events. **Quantitative forecasting** uses historical data and mathematics.

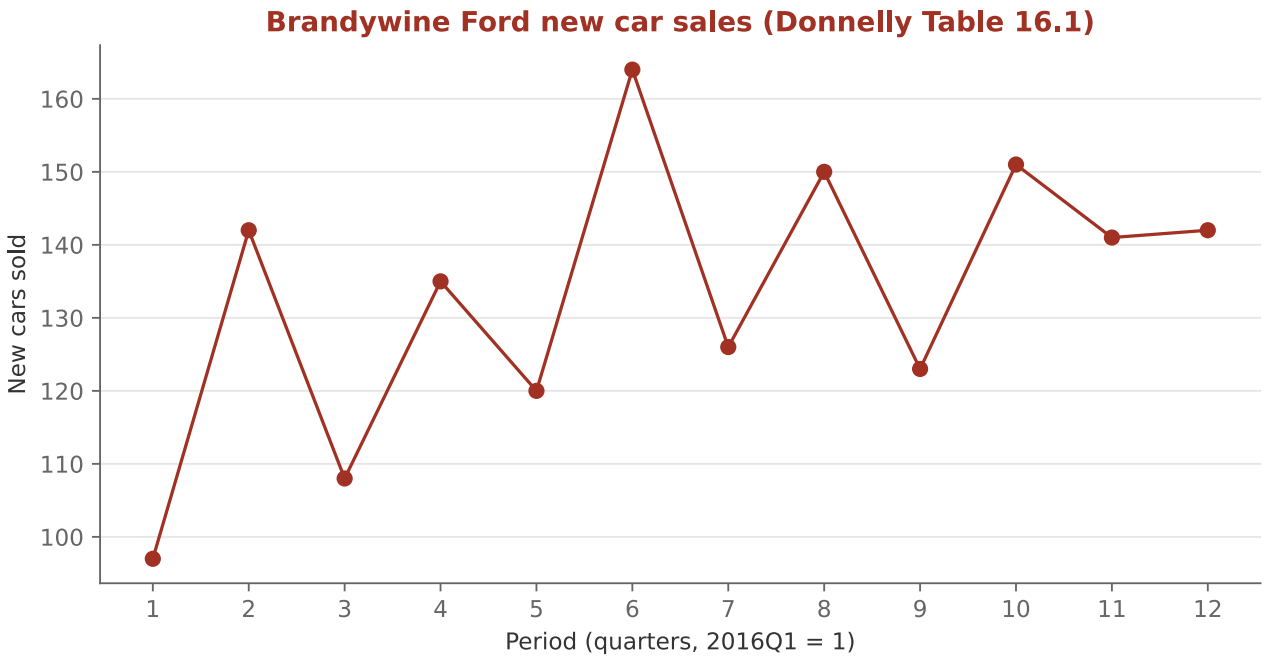
Every business decision that takes time to carry out — hiring, ordering inventory, building a plant, scheduling staff for a convention weekend — is a bet on a number nobody knows yet. A forecast is the number you bet on, and a forecasting *method* is a rule for producing it that you could apply again next period. The second half of that sentence is what separates a method from a guess.

> A **time series** is a group of data values, each associated with a specific point in time.

Donnelly's running example is Brandywine Ford, a dealership that wants to forecast next quarter's new-car sales from the last twelve quarters (Table 16.1). We will use it throughout, because you can check every number against the book. Then we will do the same thing to the CPI.

	2016Q1	2016Q2	2016Q3	2016Q4	2017Q1	2017Q2	2017Q3	2017Q4	2018Q1	2018Q2	2018Q3	2018Q4
sales	97	142	108	135	120	164	126	150	123	151	141	142

The first thing a forecaster does is *plot the series*. A table hides what a picture shows.



The Components of a Time Series

A **trend component** is present if a general upward or downward movement can be seen in the data.

A **seasonal component** is present if a consistent pattern can be associated with the calendar. Seasonal patterns repeat within a year, so you need data more often than annual to see them.

A **cyclical component** is movement that results from swings in the economy over several years — longer than a season.

The **random component** is whatever is left: movement that cannot be attributed to trend, season, or cycle. It is often called *noise*.

Look back at the car sales. The line drifts upward (trend), and every second and fourth quarter spikes (seasonal). Now look at the five challenge series. The CPI is almost pure trend. Initial claims are almost pure noise around a level. Visitor volume has a calendar in it — you can see March and October. The Nevada house price index trends, but not in a straight line, and it has the 2008 cycle in its long history.

> Every method in Chapter 16 is a bet about which components are present. The methods in *this* note assume there is a level and some noise, and nothing else. When that is wrong — when there is a trend or a season — they fail in a predictable direction, and you will see it happen to the CPI below. Knowing *how* a method fails is most of knowing when to use it.

Smoothing Methods

The goal of a **smoothing method** is to average out the random movements in a time series, because they are unpredictable. If the forecast is an average of recent values, the ups and downs of the random component should cancel.

Smoothing works best when the series is fairly consistent over time: no significant trend, seasonal, or cyclical component. Chapter 16 covers four smoothing methods. We take the first two now and add a fifth that the book skips, because it is the one every other method has to beat.

The Naive Forecast

> The **naive forecast** for period t is the actual value from period $t-1$:
>
> $$F_t = A_{t-1}.$$

"Next quarter will be like this quarter." It sounds like an insult to forecasting, and for many series it is remarkably hard to beat — the CPI, as you are about to see, is one of them. A method that cannot outperform the naive forecast has not earned its complexity. Always compute it first.

	actual	forecast	error	abs_error
2016Q1	97.0	NaN	NaN	NaN
2016Q2	142.0	97.0	45.0	45.0
2016Q3	108.0	142.0	−34.0	34.0
2016Q4	135.0	108.0	27.0	27.0
2017Q1	120.0	135.0	−15.0	15.0
2017Q2	164.0	120.0	44.0	44.0
2017Q3	126.0	164.0	−38.0	38.0
2017Q4	150.0	126.0	24.0	24.0
2018Q1	123.0	150.0	−27.0	27.0
2018Q2	151.0	123.0	28.0	28.0
2018Q3	141.0	151.0	−10.0	10.0
2018Q4	142.0	141.0	1.0	1.0

The naive forecast for 2019Q1 is 142 cars — the 2018Q4 figure. Notice the pattern in the error column: it alternates sign, because the series alternates high and low quarters and the naive forecast is always one step behind. That is the seasonal component punishing a method that does not know about seasons.

The Simple Moving Average

> A **simple moving average** (SMA) forecast averages the most recent p values of the series:
>
> $$\mathrm{SMA}(p, t) = \frac{A_{t-1} + A_{t-2} + \cdots + A_{t-p}}{p}.$$

Donnelly chooses $p = 3$ and forecasts the first quarter of 2019 from the last three quarters of 2018:

$$\mathrm{SMA}(p=3,\ 2019\ \mathrm{Q1}) = \frac{151 + 141 + 142}{3} = \frac{434}{3} = 144.7 \approx 145$$

Now the question your boss asks: *how good is that number likely to be?* The only honest way to answer is to pretend you had used the method all along, and measure how it did. Apply the three-period average to the history: the first forecast it can make is for period 4 (it needs periods 1–3), and so on.

	actual	forecast	error	abs_error
2016Q1	97.0	NaN	NaN	NaN
2016Q2	142.0	NaN	NaN	NaN
2016Q3	108.0	NaN	NaN	NaN
2016Q4	135.0	115.7	19.3	19.3
2017Q1	120.0	128.3	−8.3	8.3
2017Q2	164.0	121.0	43.0	43.0
2017Q3	126.0	139.7	−13.7	13.7
2017Q4	150.0	136.7	13.3	13.3
2018Q1	123.0	146.7	−23.7	23.7
2018Q2	151.0	133.0	18.0	18.0
2018Q3	141.0	141.3	−0.3	0.3
2018Q4	142.0	138.3	3.7	3.7

The mistake students make most: assigning the average of periods 1–3 to period 3. A forecast is for a period you do not yet know. The average of periods 1, 2 and 3 is the forecast *for period 4*. If you slide it back one row the method looks far better than it is, because it is being credited with knowing the value it was supposed to predict.

Measuring Accuracy: the Mean Absolute Deviation

The **mean absolute deviation** (MAD) is the average size of the forecasting error over the periods that have a forecast (Formula 16.1):

$$\text{MAD} = \frac{\sum |A_t - F_t|}{n},$$

where A_t is the actual value in period t, F_t the forecast for period t, and n the number of forecasting errors.

The absolute value matters. If the forecast is 10 too high one quarter and 10 too low the next, the average *error* is zero — which would be a lie about a method that missed by 10 every time. The average *absolute* error is 10.

$$\text{MAD}_{\text{SMA(3)}} = \frac{19.3 + 8.3 + 43.0 + 13.7 + 13.3 + 23.7 + 18.0 + 0.3 + 3.7}{9} = \frac{143.3}{9} = 15.9$$

A three-period moving average would have missed by about 15.9 cars a quarter. Whether that is acceptable is a business question, not a statistical one. Whether it is *better than the alternatives* is a statistical one, and it is answered by computing the alternatives.

What if $p = 4$?

	actual	forecast	error	abs_error
2016Q1	97.0	NaN	NaN	NaN
2016Q2	142.0	NaN	NaN	NaN
2016Q3	108.0	NaN	NaN	NaN
2016Q4	135.0	NaN	NaN	NaN

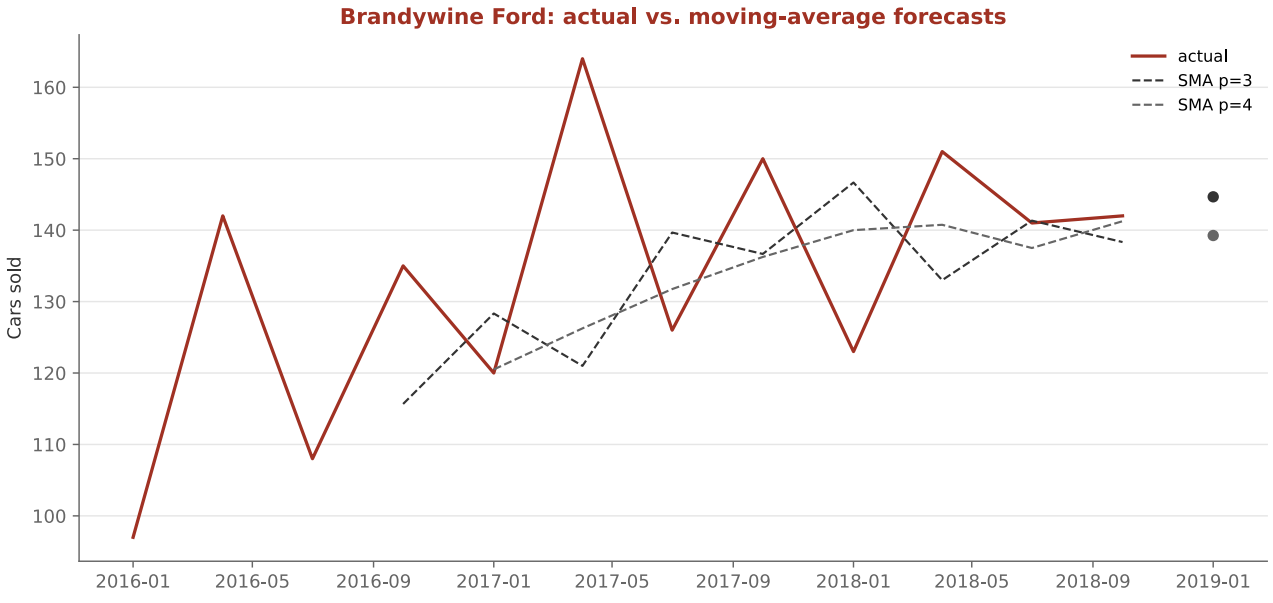
	actual	forecast	error	abs_error
2017Q1	120.0	120.5	−0.5	0.5
2017Q2	164.0	126.2	37.8	37.8
2017Q3	126.0	131.8	−5.8	5.8
2017Q4	150.0	136.2	13.8	13.8
2018Q1	123.0	140.0	−17.0	17.0
2018Q2	151.0	140.8	10.2	10.2
2018Q3	141.0	137.5	3.5	3.5
2018Q4	142.0	141.2	0.8	0.8

The four-period forecast for 2019Q1 is 139.2 with a MAD of 11.2, against 15.9 for three periods.

All else equal, choose the forecast with the lower MAD. Increasing p always smooths *more* — the forecast moves less from period to period and responds less to the latest data. Whether that lowers the error depends entirely on the series. Here it helps a lot, and it is worth asking why: a four-quarter average contains one of every season, so the seasonal spikes cancel inside the average. A three-quarter average never does. That is not a coincidence you should count on for a different series.

p	1	2	3	4	5	6
MAD	26.6	11.8	15.9	11.2	15.3	7.9
RMSE	29.7	15.2	19.9	16.0	20.1	10.4

The row for $p = 1$ is the naive forecast. Read across: the MAD does not fall smoothly as p grows. It jumps around, and the even windows win — which is the seasonal pattern again. (The $p = 6$ MAD averages only six errors; do not read too much into it.)



The dashed lines are smoother than the actual series: that is the point of the method, and also its weakness. Every peak is forecast late and low; every trough is forecast late and high. A moving average *lags*.

The Weighted Moving Average

A **weighted moving average** (WMA) applies weights to the p most recent values, conventionally giving the most recent value the largest weight. The denominator is the sum of the weights:

$$\text{WMA}(p=3,t) = \frac{3A_{t-1} + 2A_{t-2} + 1A_{t-3}}{3+2+1}.$$

The reasoning is that last month probably tells you more about next month than the month before it did. For Brandywine Ford with weights 3, 2, 1:

$$\text{WMA}(2019\ \text{Q1}) = \frac{(3)(142) + (2)(141) + (1)(151)}{6} = \frac{859}{6} = 143.2$$

	actual	forecast	error	abs_error
2016Q1	97.0	NaN	NaN	NaN
2016Q2	142.0	NaN	NaN	NaN
2016Q3	108.0	NaN	NaN	NaN
2016Q4	135.0	117.5	17.5	17.5
2017Q1	120.0	127.2	−7.2	7.2
2017Q2	164.0	123.0	41.0	41.0
2017Q3	126.0	144.5	−18.5	18.5
2017Q4	150.0	137.7	12.3	12.3
2018Q1	123.0	144.3	−21.3	21.3
2018Q2	151.0	132.5	18.5	18.5
2018Q3	141.0	141.5	−0.5	0.5
2018Q4	142.0	141.3	0.7	0.7

MAD 15.3: a little better than the three-period SMA (15.9), not as good as the four-period SMA (11.2). The SMA is a WMA with equal weights. The choice of weights is arbitrary and found by trial and error — which is a polite way of saying that if you try enough weights you will find some that fit the past well, and there is no guarantee they fit the future.

Two More Yardsticks

MAD is the textbook's measure and the one you should be able to compute by hand. Two companions are worth knowing because the challenge is scored on more than one series, and they answer slightly different questions.

The **root mean squared error** squares the errors before averaging, so one large miss counts for more than several small ones: $\text{RMSE} = \sqrt{\sum (A_t - F_t)^2 / n}$.

The **mean absolute percentage error** divides each error by the actual value, so it is unit-free: $\text{MAPE} = \frac{100}{n}\sum |A_t - F_t| / |A_t|$. A MAPE of 2% on the CPI and a MAPE of 4% on jobless claims can be compared; a MAD of 1.2 index points and a MAD of 8,000 claims cannot.

method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	142.0	26.6	29.7	19.7	11
SMA p=3	144.7	15.9	19.9	11.2	9
SMA p=4	139.2	11.2	16.0	7.6	8
WMA 3-2-1	143.2	15.3	19.2	10.8	9

The last column matters. Each method is judged on a different number of errors, because each needs a different number of periods to get started. With twelve observations that is a real limitation; with twelve *years* of monthly data it is not.

The Challenge Series: the CPI

Now a series you will actually forecast. The Consumer Price Index measures the price of a fixed basket of goods relative to 1982–84 = 100. It is monthly, seasonally adjusted, and revised rarely.

Look before you forecast

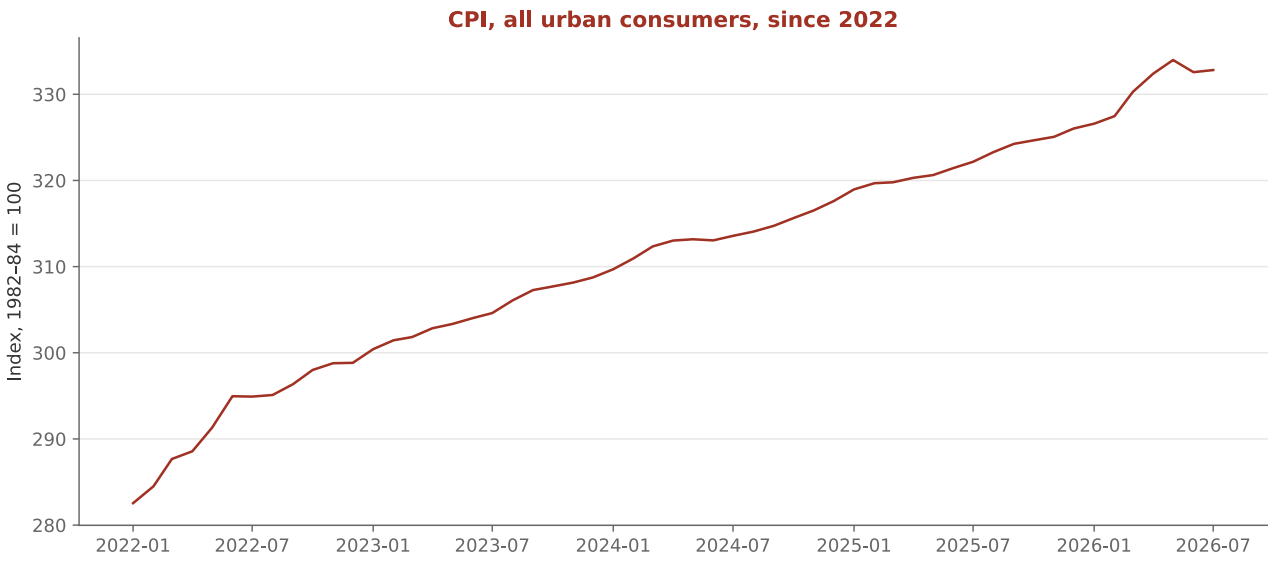
Chapter 1 taught you to check a data set before analysing it. That does not stop being true because the next step is a forecast.

	CPIAUCSL
observations	679
first	1970-01-01
last	2026-07-01
type	float64
missing	1
missing at	2025-10-01

There is a hole. The Bureau of Labor Statistics did not publish a CPI for October 2025, because the price survey was not conducted during the federal shutdown. FRED shows a `.`, and a `.` is not a zero. A moving average needs consecutive months, so we must decide what to do about it, and the decision is a modelling choice that belongs in your write-up, not a cleaning step that happens silently. Here we draw a straight line between September and November and say so.

```
NOTE: filled 1 missing value(s) by 'interpolate' at: 2025-10-01 00:00:00
```

The second decision is the window. The CPI runs back to 1947 on FRED (1970 in our file). The 1970s are real data, but they are not the regime you are forecasting. We use January 2022 onward — 55 months. How far back to look is a choice; a different window gives a different forecast, and you should know by how much.

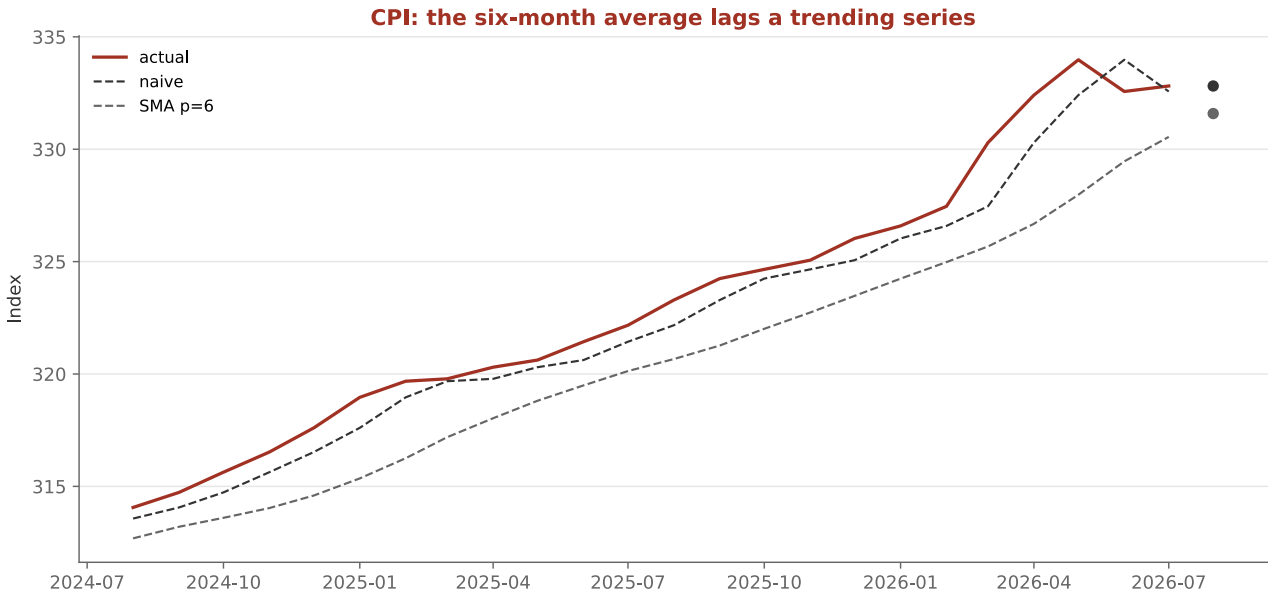


Forecast next month

method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	332.813	0.990	1.253	0.320	54
SMA p=2	332.690	1.427	1.729	0.461	53
SMA p=3	333.120	1.821	2.128	0.587	52
SMA p=6	331.587	3.013	3.209	0.963	49
WMA 1-2-3	332.926	1.513	1.790	0.487	52

Forecasts for August 2026 in the first column; the methods are then judged on every month since 2022 that has a forecast. The smallest MAD belongs to **naive**, and the ranking runs almost exactly opposite to the car-sales ranking: the widest window is the worst.

> Why? The CPI has a trend. It rises in most months, so *every average of past values sits below the current level*, and the wider the window, the further below. A six-month average is forecasting where the CPI was three months ago. Donnelly says it on p. 770: when a series has a trend or a season, smoothing methods "lag behind the data, resulting in increased forecasting errors." You have just watched it happen. The naive forecast lags by only one month, which is why it wins — for now. After Exam 1 we will meet methods that do not lag at all.



Backtest: what the challenge will do to you

The table above judges each method on the whole sample at once. The challenge does something harsher: every period it asks for one forecast, then reveals the answer. The fair test of a method is therefore to *replay* that — for each of the last 24 months, fit the method on the data available at the time, forecast one month ahead, and record the miss.

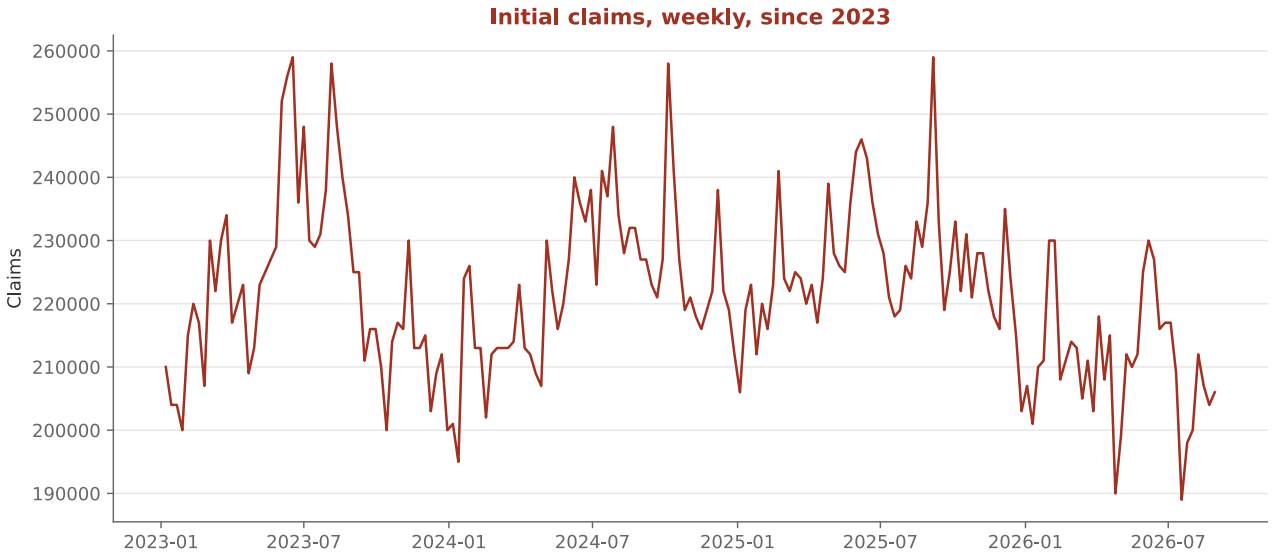
method	MAD	RMSE	MAPE	periods
naive	0.919	1.098	0.283	24
SMA p=2	1.299	1.526	0.400	24
WMA 1-2-3	1.392	1.644	0.429	24
SMA p=3	1.647	1.934	0.508	24
SMA p=6	2.825	3.047	0.870	24

Same verdict, and now it is a verdict you can defend: had you submitted the naive forecast every month for two years, you would have missed by 0.92 index points on average.

period	actual	forecast	error	abs_error	pct_error
2026-02-01	327.460	326.588	0.872	0.872	0.266
2026-03-01	330.293	327.460	2.833	2.833	0.858
2026-04-01	332.407	330.293	2.114	2.114	0.636
2026-05-01	333.979	332.407	1.572	1.572	0.471
2026-06-01	332.568	333.979	−1.411	1.411	0.424
2026-07-01	332.813	332.568	0.245	0.245	0.074

A Second Series: Weekly Initial Claims

Initial claims for unemployment insurance are released every Thursday for the week ending the previous Saturday. They are noisy, they have no trend to speak of in normal times, and they are the most frequently graded series in the challenge.



method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	206,000.0	7,684.2	10,022.6	3.4	190
SMA p=2	205,000.0	7,894.2	10,220.2	3.5	189
SMA p=4	207,250.0	8,684.5	10,847.2	3.9	187
SMA p=8	203,125.0	8,768.4	11,447.6	3.9	183
WMA 1-2-3-4	206,200.0	8,073.3	10,202.6	3.6	187

method	MAD	RMSE	MAPE	periods
SMA p=2	7,019.2	9,655.1	3.4	26
WMA 1-2-3-4	7,526.9	9,745.8	3.6	26
naive	7,692.3	9,762.6	3.7	26
SMA p=4	8,375.0	10,327.2	4.0	26
SMA p=8	8,557.7	11,513.5	4.1	26

Forecast for the week ending 2026-09-05. Over the last 26 weeks the best method is **SMA p=2**, and the worst is only 22% behind it. Claims have no trend, so averaging costs almost nothing — unlike the CPI, where every average paid for its width. Same tools, different verdict, and the difference is in the *data*, not the method.

> This is why the challenge makes you look every period rather than pick a method once in September. The MAPE column says a good week is a 3–4% miss on claims and a 0.3% miss on the CPI. When you report a forecast, report the error you expect alongside it. A number without its error is not a forecast; it is a wish.

Getting the Data

The syllabus says you should not be transcribing numbers by hand, and it means it. But there is a constraint you need to know about first.

> **Claude's code sandbox has no internet access.** Claude can run Python on files you upload; it cannot reach FRED, the LVCVA, or GitHub from inside the code it runs, whatever your account tier. So every route below ends the same way: *you* download a file with one click, and Claude reads it. A script that "downloads the data" is for your own laptop, not for Claude — and if Claude ever hands you a series it did not read from a file you gave it, it made the series up.

FRED: the Download button

Open the series page — https://fred.stlouisfed.org/series/CPIAUCSL — click **Download**, choose **CSV**. You get a two-column file, `observation_date` and `CPIAUCSL`, with a `.` wherever a value is missing. Upload it to Claude alongside `_forecast.py` and the script you are working from. That is the whole procedure, for all four FRED series, every week.

If you have Python on your own computer, `scripts/fred_download.py` does the same thing from the command line and writes a provenance file next to the CSV:

```
python fred_download.py CPIAUCSL ICSA HOUST NVSTHPI
```

If you have a free FRED API key, set it once as the environment variable `FRED_API_KEY` and the script will use the `fredapi` package as a fallback. **Never paste the key into a script, a notebook, or a chat.**

Las Vegas visitor volume: the LVCVA workbook

Visitor volume is not on FRED. The LVCVA posts an Excel workbook for each year on its research page: a **Year-to-Date Summary** for the current year and a **Year-End Summary** for each past year. Row 8 of the first sheet is Visitor Volume, one cell per month, with an `r` beside any month that has been revised.

`scripts/lvcva_tools.py` — the Python version of the R utilities this course used before Fall 2026 — reads those workbooks and writes `lv_visitors.csv`. The procedure inside Claude:

1. On the research page, download the Year-to-Date Summary and the Year-End Summaries for 2022 through last year (four or five files, a click each).
2. Start a Claude conversation. Upload the workbooks, `lvcva_tools.py`, and `COURSE_CONTEXT.md`. All three are in the **data-tools pack**, together with a copy of `lv_visitors.csv` as of the day the note was built, in case the LVCVA page is down.
3. Paste this:

> **PROMPT — BUILD THE VISITOR-VOLUME SERIES FROM THE LVCVA WORKBOOKS**
>
> I have uploaded several LVCVA "Summary of Monthly Tourism Indicators" workbooks and the course file `lvcva_tools.py`. Run
>
> ```
> python lvcva_tools.py <the workbook file names>
> ```
>
> and show me its output verbatim. It should report the year and the number of published months it found in each workbook, then write `lv_visitors.csv`. Show me the last six rows of that file and the first three, and tell me the date of the newest month.
>
> If the script errors on a workbook, show me the traceback and stop. Do not type the numbers in from the workbook yourself, and do not fill in a month that is not there.

The newest month in the file is the one the LVCVA has published; the challenge asks you for the one after it. Keep the workbooks: when next month's summary appears, download the new Year-to-Date file, rerun the same command, and the series grows by one row — and possibly changes in the row before, because the LVCVA revises.

On your own computer with Python and internet, `python lvcva_tools.py` with no arguments finds and downloads the workbooks itself. It pauses a few seconds between files so as not to hammer a public server; leave the pause in.

What is in the course packs

Each lab pack ships the CSVs the note was built from. They are a snapshot: correct on the day they were made and one release behind by the time you read this. Use them to *learn the method*; use a fresh download to *submit a forecast*. The `data/README.md` in the repository records the retrieval date of every file.

Working with Claude

The Rebel Spark initiative gives you Claude with the full tool suite. That changes what can be expected of you: the clicking is no longer the skill. The *specification* is the skill — which series, which window, which method, what to do about the hole in October 2025 — and so is the *verification*. Below are prompts that work, the reason they work, and the checks you owe before you believe the answer.

The lab prompt

RUN THIS WITH CLAUDE

> Get the MBA 775 course materials from https://github.com/jcrooker/mba775. I need `scripts/16a_naive_sma_wma.py`, `scripts/_course.py`, `scripts/_forecast.py`, `data/cpi.csv` and `data/icsa.csv`. Read `COURSE_CONTEXT.md` from the same repository first — it explains how this course expects analytical work to be done.
>
> Run the script and show me its actual output, verbatim. Then walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot reach GitHub, tell me plainly and I will upload the files instead.

A bad prompt and a good one

Bad: *"Forecast the CPI for next month."* You will get a number. You will not know what data it came from, what method produced it, or whether Claude computed it or remembered a headline.

Good — it names every decision this note asked you to make:

PROMPT — A FORECAST YOU CAN DEFEND

> I have uploaded `cpi.csv` (FRED CPIAUCSL, monthly) and `_forecast.py` from my course. Using the data from January 2022 onward, and interpolating the single missing month (October 2025) — say so in the output — produce a forecast for the next month by each of: naive, simple moving averages with p = 2, 3 and 6, and a weighted moving average with weights 1-2-3 (most recent weighted 3).
>
> For each method report the next-month forecast, the MAD, RMSE and MAPE over the sample, and a rolling-origin backtest over the last 24 months using `backtest_summary`. Show the code you ran and its output verbatim. Then tell me which method has the lowest backtested MAD and why the wide-window moving average does worse on this particular series.

> Every number must come from executed code. If a function is missing or errors, show me the traceback.

Verify before you submit

Six checks, each under a minute:

Check	What it catches
The last date printed is the month you think it is	A stale file, or a series that has not updated
The forecast is for the *next* period, not the last one	The period-3/period-4 mistake, at scale
The MAD was averaged over the right number of errors	A method credited with forecasts it never made
Recompute one forecast by hand from the last three values	A method that is not the one you asked for
The missing month was mentioned, not silently dropped	A moving average that skipped a month without saying
The number in the prose matches the number in the output	The single most common failure: confident prose, different number

Show it fail first. Try the bad prompt once, on purpose, and run the checklist against what comes back. The failures are predictable: a forecast for the wrong period, a figure quoted from memory, a moving average with a different window from the one you named, a missing month filled without comment. A student who has caught one of these will check for the rest of their career. A student handed a working answer has learned nothing.

Have Claude argue with you

PROMPT — A SECOND OPINION ON YOUR CHOICE

> Here is the backtest table my script produced for weekly initial claims (paste it). I plan to submit the SMA p = 2 forecast. Give me the strongest case *against* that choice using only the numbers in the table, then tell me what I would need to see over the next four weeks to change my mind. Do not compute new numbers; argue from these.

Excel, if you want it

PROMPT — A FORECASTING WORKBOOK

> Build an Excel workbook from `cpi.csv`. Sheet `Data`: the raw file, headers in row 1. Sheet `Forecast`: columns for date, actual, naive forecast, 3-month SMA, 3-2-1 WMA, and the absolute error of each, using live formulas that reference the `Data` sheet — no pasted values. The first SMA row must be the fourth month, not the third. Put the MAD of each method in a labelled cell at the top, computed with `AVERAGE` over the error column, and a line chart of actual versus each forecast. Tell me which cells hold formulas and which hold constants.

When it opens, change one CPI value and watch which forecasts and MADs move. The ones that do not are pasted values, and the workbook is wrong.

Your Turn (from the textbook)

A student wants to forecast her GPA for the fall of her senior year from nine semesters on a trimester schedule (Donnelly, Your Turn 1 and 2, p. 769–770):

	Fr Fall	Fr Spring	Fr Summer	So Fall	So Spring	So Summer	Jr Fall	Jr Spring	Jr Summer
GPA	2.50	2.20	2.70	3.00	2.60	3.00	3.20	2.70	3.30

(a) Calculate a two-period SMA forecast for the fall semester of her senior year and its MAD. (b) Do the same with a two-period WMA with weights 2 and 1.

(b) Which is the better forecast? Do it by hand first; then run the code by unfolding the block below and check yourself.

method	next_forecast	MAD	RMSE	MAPE	n_errors
SMA p=2	3.00	0.36	0.37	12.21	7
WMA 2-1	3.10	0.38	0.38	12.91	7

Laboratory 16a

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 16a pack — one file. Unzip it and you will have a folder holding 6 files. Drag all 6 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

I have uploaded 6 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

Then run `16a_naive_sma_wma.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

PROMPT

Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 16a and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

Then run `16a_naive_sma_wma.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach GitHub, **nothing is wrong with your work or with the repository.** Web access is a setting, and even when it is on Claude cannot always retrieve a file by direct link. Use Path 1.

Your first submission

Once the script runs, you have everything you need for this week's challenge entries: a forecast, a method, a window, and a measured error. Submit the forecast in Canvas and keep the script. When the number is published, rerun it — the new observation goes on the end of the series, the backtest gets one row longer, and you find out whether your method is still the one you would choose.

What to write down with every submission (a sentence each): the series and its last observation date; the method and its parameters; the window of data; the backtested MAD; and the one thing you would want to know before trusting the forecast. That is the whole discipline of forecasting, and it fits on an index card.